

16x16 2D Systolic Array Hardware Accelerator

Vivado 2025.1 ML Edition: Complete Simulation, Synthesis & Hardware Guide

Target Hardware: TUL PYNQ-Z2 (Zynq-7000 xc7z020clg400-1)
Baseline Board: Digilent Basys 3 (Artix-7 xc7a35tcpg236-1)

Array Scale: 16x16 Mesh (256 Processing Elements)
Precision: INT8 Inputs (Signed) -> INT16 Accumulation
Core Frequency: 125 MHz (PYNQ-Z2) / 100 MHz (Basys 3)

1. Project Architecture & File Classification

The project is structured into modular, self-contained directories. Each target board contains three distinct subfolders: **board_implementation/** (board wrapper and constraints), **src/** (core synthesizable modules), and **sim/** (testbenches).

Category	Folder Path	Key Files Included	Purpose / Hardware Role
PYNQ-Z2 Board	boards/pynq_z2/board_implementation/	pynq_z2_demo_16x16_top.sv pynq_z2_constraints.xdc	Standalone hardware demo top with ROM test matrices, LEDs, RGB LEDs, buttons, and 115200 baud UART stream.
Basys 3 Board	boards/basys3/board_implementation/	basys3_demo_16x16_top.sv basys3_constraints.xdc seven_segment_ctrl.sv	Basys 3 hardware demo with 7-segment display showing 16-bit hex matrix elements.
Core RTL	rtl/ or boards/*/src/	systolic_pkg.sv, processing_element.sv, skew_buffer.sv, systolic_array.sv, controller.sv, systolic_top.sv	Synthesizable hardware accelerator modules. Top module is systolic_top.sv .
Testbenches	tb/unit/ or boards/*/sim/	tb_step6_systolic_16x16.sv	Self-checking unit testbench verifying Identity ($A \cdot I = A$), $2 \cdot 1$, and dense signed matrix multiply (768 checks).
OOP Testbench	tb/oop/ or boards/pynq_z2/sim/	tb_matrix_top.sv, MatrixGenerator.sv, MatrixScoreboard.sv, MatrixDriver.sv, MatrixMonitor.sv	Enterprise randomized UVM-style verification running 30 full matrix transactions (7,680 checks).

2. Vivado 2025.1 Project Setup (GUI Mode)

Follow these steps to create and configure a clean project in Vivado 2025.1 ML Edition without any Tcl scripts:

Step	Action in Vivado GUI	Exact Selection / File Path
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1	Create New Project	Open Vivado -> Quick Start: Create Project -> Name: <i>systolic_16x16</i> -> Select RTL Project (Do not specify sources now).
2	Select Target FPGA Part	In Default Part, select: • For PYNQ-Z2: xc7z020clg400-1 • For Basys 3: xc7a35tcpg236-1
3	Add Synthesizable RTL Sources	Flow Navigator -> Add Sources -> <i>Add or create design sources</i> -> Browse to boards/pynq_z2/src/ and select all .sv and .vh files.
4	Add Board Implementation Top	In Add Design Sources, also add boards/pynq_z2/board_implementation/pynq_z2_demo_16x16_top.sv .
5	Add Physical Constraints (XDC)	Add Sources -> <i>Add or create constraints</i> -> Add boards/pynq_z2/board_implementation/pynq_z2_constraints.xdc .
6	Add Simulation Testbenches	Add Sources -> <i>Add or create simulation sources</i> -> Add boards/pynq_z2/sim/tb_step6_systolic_16x16.sv (or all .sv in sim/).

3. Top Module Selection Guide

Vivado requires setting different Top Modules depending on whether you are running a **Simulation** or generating a **Hardware Bitstream**. Here is the definitive reference table:

Target Task	Top Module Name	File Path to Select	Action in Vivado
16x16 Functional Simulation	tb_step6_systolic_16x16	boards/pynq_z2/sim/tb_step6_systolic_16x16.sv	In Sources -> Simulation Sources , right-click tb_step6_systolic_16x16.sv and select Set as Top . Click <i>Run Behavioral Simulation</i> .
OOP 30-Transaction Verification	tb_matrix_top	boards/pynq_z2/sim/tb_matrix_top.sv	In Simulation Sources , right-click tb_matrix_top.sv and select Set as Top . Click <i>Run Behavioral Simulation</i> .
PYNQ-Z2 Bitstream Synthesis	pynq_z2_demo_16x16_top	boards/pynq_z2/board_implementation/pynq_z2_demo_16x16_top.sv	In Sources -> Design Sources , right-click pynq_z2_demo_16x16_top.sv and select Set as Top . Click <i>Generate Bitstream</i> .
Basys 3 Bitstream Synthesis	basys3_demo_16x16_top	boards/basys3/board_implementation/basys3_demo_16x16_top.sv	In Design Sources , right-click basys3_demo_16x16_top.sv and select Set as Top . Click <i>Generate Bitstream</i> .

4. Waveform & Timing Analysis

When running behavioral simulation in Vivado XSim, the 16x16 accelerator exhibits strict deterministic timing:

Key Waveform Signals to Monitor:

- **clk** : 100 MHz clock (10 ns period) in testbench, 125 MHz on PYNQ-Z2.
- **rst** : Active-high synchronous reset pulse (asserted for 2 cycles).
- **start** : 1-clock-cycle pulse triggering state transition from IDLE to LOAD.
- **load_en** : High for exactly 1 cycle while parallel matrices are latched into skew buffers.
- **shift_en & array_en** : High for $(3N - 1) = 47$ clock cycles while diagonal wavefront feeds the 256 PEs.
- **done** : Asserted exactly at cycle 48 (500 ns @ 100 MHz / 384 ns @ 125 MHz).
- **result[15:0][15:0]** : Holds the complete 16x16 signed matrix C when done is asserted.

5. PYNQ-Z2 Hardware Demonstration Guide

Once the bitstream is generated and programmed to your PYNQ-Z2 board via Vivado Hardware Manager, interact with the physical board as follows:

Control / Indicator	Pin / Color	Hardware Function & Meaning
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Slide Switches sw[1:0]	Pins M19, M20	00: Matrix A * Identity ($C = A$) 01: Matrix A * 2*Identity ($C = 2A$) 10: Dense Signed INT8 GEMM 11: All-Ones Test ($C[i][j] = 16$)
btn[1] (START)	Pin D20	Press once to initiate 16x16 matrix multiplication across all 256 PEs.
btn[0] (RESET)	Pin D19	Synchronous reset clears internal registers and latches.
led[0] (Heartbeat)	Pin R14	Smoothly blinks at ~1.86 Hz to verify the 125 MHz clock and FPGA logic are active.
led[1] (Busy)	Pin P14	High during the 48 cycles of active systolic compute.
led[2] (DONE)	Pin N16	Lights up solid when the full 16x16 matrix is calculated.
led[3] (PASS)	Pin M14	256/256 Hardware Verification PASS indicator. Lights up solid when all elements match expected golden results.
rgbled4 (Core State)	Pins M15, T16, Q15	Blue: IDLE / Ready Amber: COMPUTING Vibrant Green: DONE & 100% VERIFIED
UART Serial Output	Pin Y18 (PMOD A1)	Connect USB-UART adapter at 115200 Baud, 8N1 . Prints 16x16 GEMM: PASS upon completion.